1.9 Exercises

Exercise 1.1 – Define and draw a function

(a) Open the ROOT interpreter.

(b) Create a one-dimensional function (`TF1`) representing the parabola

$$f(x) = x^2 + 2x + 1,$$

defined in the range $[-10, 10]$.

(c) Change the line color and the line style, then draw the function.

(d) Before drawing, predict the position of the minimum of f and its value. Check your prediction with `GetMinimumX()` and `GetMinimum()`.

(e) Rewrite the function with parameters, $f(x) = [0]\,x^2 + [1]\,x + [2]$, set the parameters to $(1, 2, 1)$ and check that you obtain the same curve. Then change [0] to -1: draw the new function on the same canvas (option `"SAME"`) with a different color.

Exercise 1.2 – Define and draw a cubic function

Write a ROOT macro called `drawCubic.C` that answers the following questions.

(a) Define a function representing the cubic

$$g(x) = x^3 - 3x^2 - 24x + 5$$

in the range $[-7, 7]$.

(b) Change the line style, the line color and the line width.

(c) Create a canvas and draw the function on it.

(d) Add a title and axis labels (e.g. with `g->SetTitle("Cubic;x;g(x)")`), then save the canvas as `cubic.png` and `cubic.pdf`.

(e) Compute $g'(x)$ by hand and find the positions of the local maximum and minimum. Check them with `GetMaximumX()` and `GetMinimumX()`, choosing appropriate sub-ranges.

Instructions: use `TF1` to define the function; use `SetLineColor()`, `SetLineStyle()` and `SetLineWidth()` for the style; use `TCanvas` to create the canvas, `Draw()` to plot the function and `SaveAs()` to save it.

Exercise 1.3 – Fit a histogram with an exponential function

(a) Open the ROOT interpreter.

(b) Create a histogram with 80 bins in the range $[0, 10]$. What do the number of bins and the range mean? What is the width of one bin?

(c) Fill it with 10 000 random numbers distributed according to the decreasing exponential $e^{-x/2}$. *Hint:* define a generating function `TF1 gen("gen", "exp(-0.5*x)", 0, 10)` and use `h.FillRandom("gen", 10000)`.

(d) Define the exponential fit function

$$f(x) = [0] \cdot e^{-[1]\,x},$$

where [0] and [1] are the fit parameters. Give them reasonable initial values with `SetParameters()`. What value do you expect for [1]?

(e) Fit the histogram with this function.

(f) Print the fit parameters and their uncertainties. Is [1] compatible with its true value within its uncertainty?

(g) Draw the histogram together with the fit result. Switch to a logarithmic y axis (`gPad->SetLogy()`): what does an exponential look like on this scale, and why?

(h) ROOT also has a predefined function `"expo"`, equal to $e^{p_0+p_1 x}$. Fit the histogram with it and relate p_0 and p_1 to the parameters [0] and [1] of your own function.

Exercise 1.4 – Gaussian fit in a macro

Write a macro, called `myFit.C`, that answers the following questions.

(a) Create a histogram with 100 bins between -5 and $+5$.

(b) Fill the histogram with 10 000 random numbers drawn from a Gaussian distribution of mean $\mu = 0$ and standard deviation $\sigma = 1$. What shape do you expect for the histogram?

(c) Define a Gaussian function of the form

$$f(x) = A \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right].$$

What do the parameters A, μ and σ represent? Using the number of entries and the bin width, predict the value of A.

(d) Fit the histogram with the Gaussian function.

i. What values do you obtain for μ and σ?

ii. Are they close to the true ones (0 and 1)? Express the difference in units of the fit uncertainty.

iii. Why might they not be exactly the same?

(e) Draw the histogram and the fitted function together, and display the fit results on the plot with `gStyle->SetOptFit(1111)`.

(f) How can you check that the fit describes the data well? Read the χ^2 and the number of degrees of freedom (`GetChisquare()`, `GetNDF()`) and comment on their ratio.

(g) Repeat the fit with 100 entries instead of 10 000. How do the uncertainties on μ and σ change? By which factor did you expect them to change?

(h) With 100 entries, many bins are empty or contain very few events. Repeat the fit with the likelihood option `h->Fit(f, "L")` and compare with the χ^2 fit. Which one should you trust here, and why?

Exercise 1.5 – Save and read back the fit result

(a) At the end of `myFit.C`, open a file `fit.root` with the option `"RECREATE"` and write the histogram and the fitted function into it. Close the file.

(b) List the content of the file with `rootls -l fit.root` and with a `TBrowser`.

(c) In a new ROOT session, open the file, retrieve the histogram and the function with `Get<TH1D>()` and `Get<TF1>()`, draw them again and print the fitted value of σ.

(d) Do the same from Python with `import ROOT`. Then read the histogram with `uproot` and print its bin contents with `.values()`.

(e) What happens to `fit.root` if you run `myFit.C` again? Which option would you use to add a second histogram to the existing file without erasing it?